

llama.cpp LLM模型Windows CPU 部署（运行LLama2测试）

cpu: AMD Ryzen 7 5800 8-Core Processor

LLAMA2-7B的参数量化过后大约13GB，模型运行后cpu利用率75%左右。

准备

安装MinGW

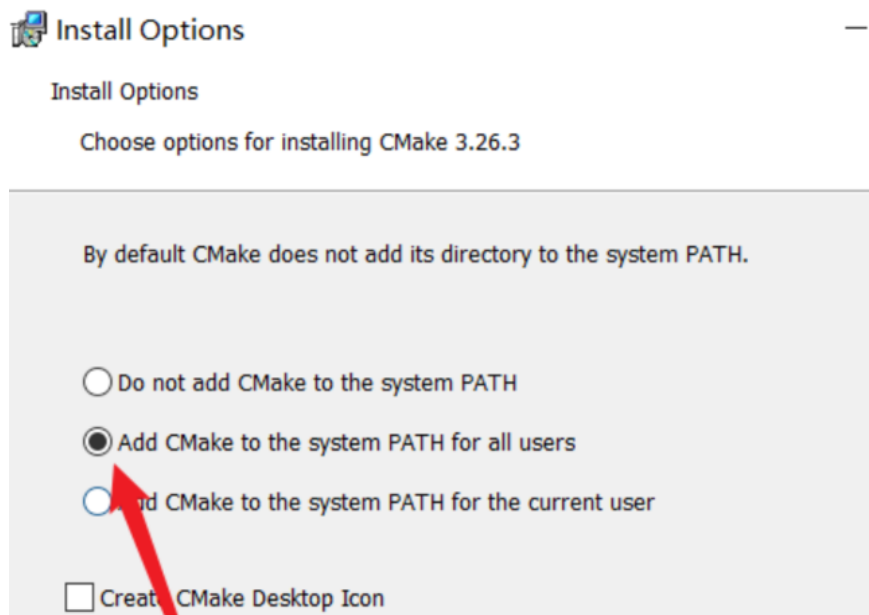
llama.cpp使用c++语言写的，因此安装编译器MinGW（Windows版g++编译工具）

安装之后需要把安装路径添加到系统环境变量，运用gcc -v查看是否安装成功

安装Cmake

下载地址: <https://cmake.org/download/>

下载安装，环境变量可以安装软件步骤中选择自动加入（如下图），也可以自己手动加入到系统变量中



安装完成之后如下图

```
C:\Users\loong>cmake
Usage

  cmake [options] <path-to-source>
  cmake [options] <path-to-existing-build>
  cmake [options] -S <path-to-source> -B <path-to-build>

Specify a source directory to (re-)generate a build system for it in the
current working directory. Specify an existing build directory to
re-generate its build system.

Run 'cmake --help' for more information.
```

llama.cpp下载编译

下载

创建项目文件夹

在该项目文件夹下使用git下载

```
1 git clone https://github.com/ggerganov/llama.cpp
2 cd llama.cpp
3 ls
```

cmake编译

```
1 mkdir build
2 cd build
3 cmake ..
4 cmake --build . --config Release
```

注意：

1.如果 `cmake ..` 编译出现not set 问题，请使用 `cmake .. -G "MinGW Makefiles"` 重新编译

2.若该命令编译失败：在CMakeLists文件中设置了以下参数重新编译

```
#SET(CMAKE_MAKE_PROGRAMC "D:/Soft/mingw64/bin/mingw32-make.exe")
#SET(CMAKE_C_COMPILER "D:/Soft/mingw64/bin/gcc.exe")
#SET(CMAKE_CXX_COMPILER "D:/Soft/mingw64/bin/g++.exe")
```

3.若出现下图错误，请将编译记录删除：可以直接删除项目，重新下载

```
PS D:\Work\llama2\llama.cpp\build> cmake .. -G "MinGW Makefiles"
CMake Error: Error: generator : MinGW Makefiles
Does not match the generator used previously: NMake Makefiles
Either remove the CMakeCache.txt file and CMakeFiles directory or choose a different binary directory.
```

测试运行

```
1 在bin目录下
2 ./main -h
```

运行LLaMA-7B模型测试

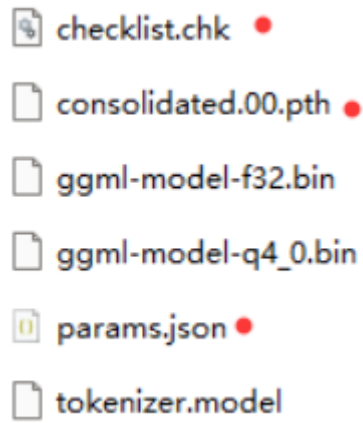
模型下载

<https://huggingface.co/nyanko7/LLaMA-7B/tree/main>

请求链接[Llama access request form - Meta AI](#)

具体申请步骤可以参考：【Llama2模型申请与本地部署详细教程】 https://www.bilibili.com/video/BV1H14y1X7kD/?share_source=copy_web&vd_source=c52fdad139a3f6e2bf4086a79c9575ad

文件down后如图所示有三个，下载下来的源码放到\llama.cpp\models\7B



注意：tokenizer.model本身下载参数里面没有，需要从源码手动拉一个

链接：<https://github.com/facebookresearch/llama.git>

转换为ggml F32 格式

- 1 PS D:\Work\llama2\llama.cpp> python convert.py .\llama.cpp\7B\
- 2
- 3 注：convert.py文件就在llama.cpp下

```
PS D:\Work\llama2\llama.cpp> python convert.py models/7B/
Loading model file models\7B\consolidated.00.pth
vocabtype: spm
Loading vocab file models\7B\tokenizer.model
params: n_vocab:32000 n_embd:4096 n_mult:256 n_head:32 n_layer:32
Writing vocab...
[ 1/291] Writing tensor tok_embeddings.weight | size 32000 x 4096 | type UnquantizedDataType(name='F32')
[ 2/291] Writing tensor norm.weight | size 4096 | type UnquantizedDataType(name='F32')
[ 3/291] Writing tensor output.weight | size 32000 x 4096 | type UnquantizedDataType(name='F32')
[ 4/291] Writing tensor layers.0.attention.wq.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[ 5/291] Writing tensor layers.0.attention.wk.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[ 6/291] Writing tensor layers.0.attention.wv.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[ 7/291] Writing tensor layers.0.attention.wo.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[ 8/291] Writing tensor layers.0.attention_norm.weight | size 4096 | type UnquantizedDataType(name='F32')
[ 9/291] Writing tensor layers.0.feed_forward.w1.weight | size 11008 x 4096 | type UnquantizedDataType(name='F32')
[10/291] Writing tensor layers.0.feed_forward.w2.weight | size 4096 x 11008 | type UnquantizedDataType(name='F32')
[11/291] Writing tensor layers.0.feed_forward.w3.weight | size 11008 x 4096 | type UnquantizedDataType(name='F32')
[12/291] Writing tensor layers.0.ffn_norm.weight | size 4096 | type UnquantizedDataType(name='F32')
[13/291] Writing tensor layers.1.attention.wq.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[14/291] Writing tensor layers.1.attention.wk.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[15/291] Writing tensor layers.1.attention.wv.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[16/291] Writing tensor layers.1.attention.wo.weight | size 4096 x 4096 | type UnquantizedDataType(name='F32')
[289/291] Writing tensor layers.31.feed_forward.w2.weight | size 4096 x 11008 | type UnquantizedDataType(name='F32')
[290/291] Writing tensor layers.31.feed_forward.w3.weight | size 11008 x 4096 | type UnquantizedDataType(name='F32')
[291/291] Writing tensor layers.31.ffn_norm.weight | size 4096 | type UnquantizedDataType(name='F32')
Wrote org-models\ggml-model-f32.bin
```

量化

- 1 build\bin\quantize.exe D:\work\llama2\llama.cpp\models\7B\ggml-model-f32.bin D:\work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin q4_0

注：ggml-model-q4_0.bin已有资源。链接如下：

<https://huggingface.co/TheBloke/Llama-2-7B-Chat-GGML>

<https://huggingface.co/TheBloke/Llama-2-13B-chat-GGML>

```

D:\Work\llama2\llama.cpp\build\bin\quantize.exe D:\Work\llama2\llama.cpp\models\7B\ggml-model-f32.bin D:\Work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin q4_0
main: build = 1809 (9e232ffe)
main: quantizing 'D:\Work\llama2\llama.cpp\models\7B\ggml-model-f32.bin' to 'D:\Work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin' as Q4_0
llama.cpp: loading model from D:\Work\llama2\llama.cpp\models\7B\ggml-model-f32.bin
llama.cpp: saving model to D:\Work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin
[ 1/ 291]      tok_embeddings.weight = 4896 x 32808, type = f32, quantizing to q4_0 .. size = 500.00 MB -> 78.31 MB | hist: 0.037 0.016 0.025 0.039 0.057 0.077
96 1.110 0.116 0.111 0.096 0.077 0.057 0.039 0.025 0.021      4896, type = f32, size = 0.016 MB
[ 2/ 291]      norm.weight = 4896, type = f32, size = 0.016 MB
[ 3/ 291]      output.weight = 4896 x 32808, type = f32, quantizing to q4_0 .. size = 500.00 MB -> 102.54 MB | hist:
[ 4/ 291]      layers.0.attention.wq.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.034 0.008 0.012 0.019 0.031 0.050
94 0.149 0.256 0.159 0.084 0.051 0.031 0.019 0.012 0.010      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.034 0.008 0.013 0.021 0.033 0.054
[ 5/ 291]      layers.0.attention.wk.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.034 0.008 0.013 0.021 0.033 0.054
99 0.158 0.226 0.151 0.089 0.054 0.033 0.021 0.013 0.011      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.015 0.024 0.036 0.053 0.074
[ 6/ 291]      layers.0.attention.wv.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.015 0.024 0.036 0.053 0.074
96 1.117 0.129 0.117 0.096 0.074 0.053 0.036 0.024 0.020      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.035 0.011 0.017 0.028 0.044 0.068
[ 7/ 291]      layers.0.attention.wo.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.035 0.011 0.017 0.028 0.044 0.068
98 0.135 0.155 0.135 0.100 0.068 0.044 0.028 0.017 0.014      4896, type = f32, size = 0.016 MB
[ 8/ 291]      layers.0.attention_norm.weight = 4896, type = f32, size = 0.016 MB
[ 9/ 291]      layers.0.feed_forward.w1.weight = 4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.057 0.077
96 1.111 0.117 0.111 0.096 0.077 0.057 0.039 0.025 0.021      4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.036 0.015 0.025 0.038 0.056 0.076
[ 10/ 291]      layers.0.feed_forward.w2.weight = 11088 x 4896, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.036 0.015 0.025 0.038 0.056 0.076
97 1.112 0.118 0.112 0.097 0.077 0.056 0.038 0.025 0.021      4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.056 0.076
[ 11/ 291]      layers.0.feed_forward.w3.weight = 4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.056 0.076
96 1.111 0.117 0.111 0.096 0.077 0.056 0.038 0.025 0.021      4896, type = f32, size = 0.016 MB
[ 12/ 291]      layers.0.feed_forward.w4.weight = 4896, type = f32, size = 0.016 MB
[ 13/ 291]      layers.1.attention.wq.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.013 0.022 0.034 0.051 0.074
98 0.121 0.132 0.121 0.098 0.074 0.052 0.034 0.022 0.018      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.013 0.022 0.034 0.051 0.074
[ 14/ 291]      layers.1.attention.wk.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.013 0.022 0.034 0.051 0.074
99 0.121 0.132 0.121 0.099 0.074 0.051 0.034 0.022 0.018      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.014 0.023 0.035 0.052 0.073
[ 15/ 291]      layers.1.attention.wv.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.014 0.023 0.035 0.052 0.073
97 0.119 0.130 0.119 0.097 0.074 0.052 0.035 0.023 0.019      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.035 0.013 0.022 0.033 0.051 0.072
[ 16/ 291]      layers.1.attention.wo.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.035 0.013 0.022 0.033 0.051 0.072
98 0.121 0.132 0.121 0.098 0.074 0.052 0.034 0.022 0.018      4896, type = f32, size = 0.016 MB
[ 17/ 291]      layers.1.attention_norm.weight = 4896, type = f32, size = 0.016 MB
[ 18/ 291]      layers.1.feed_forward.w1.weight = 4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.057 0.077
96 1.111 0.117 0.111 0.096 0.077 0.057 0.039 0.025 0.021      4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.036 0.015 0.025 0.038 0.056 0.076
[ 19/ 291]      layers.1.feed_forward.w2.weight = 11088 x 4896, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.036 0.015 0.025 0.038 0.056 0.076
97 1.112 0.118 0.112 0.097 0.077 0.056 0.038 0.025 0.021      4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.056 0.076
[ 20/ 291]      layers.1.feed_forward.w3.weight = 4896 x 11088, type = f32, quantizing to q4_0 .. size = 172.00 MB -> 24.19 MB | hist: 0.037 0.016 0.025 0.039 0.056 0.076
96 1.111 0.117 0.111 0.096 0.077 0.056 0.038 0.025 0.021      4896, type = f32, size = 0.016 MB
[ 21/ 291]      layers.1.feed_forward.w4.weight = 4896, type = f32, size = 0.016 MB
[ 22/ 291]      layers.2.attention.wq.weight = 4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB -> 9.00 MB | hist: 0.036 0.013 0.022 0.034 0.051 0.074
98 0.121 0.132 0.121 0.098 0.074 0.052 0.034 0.022 0.018      4896 x 4896, type = f32, quantizing to q4_0 .. size = 64.00 MB ->
```

命令行交互

```
1 .\build\bin\main.exe -m D:\work\llama2\llama.cpp\models\7B\ggml-model-
  q4_0.bin -n 128 --repeat_penalty 1.0 --color -i -r "User:" -f
  D:\work\llama2\llama.cpp\prompts\chat-with-bob.txt
2
3
4 ggml-model-f32.bin
```

```

C:\Users\user> cd D:\Work\llama2\llama.cpp -> .\build\bin\main.exe -D D:\Work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin -n 128 --repeat_penalty 1.0 --color -i -r "User:" -f D:\Work\llama2\llama.cpp\prompts\chat-with-bob.txt
main: build = 1009 (9e232f0)
main: seed = 1692097241
llama.cpp: loading model from D:\Work\llama2\llama.cpp\models\7B\ggml-model-q4_0.bin
llama_model_load_internal: format = ggjt v3 (latest)
llama_model_load_internal: n_vocab = 32000
llama_model_load_internal: n_ctx = 512
llama_model_load_internal: n_embd = 4096
llama_model_load_internal: n_mult = 256
llama_model_load_internal: n_head = 32
llama_model_load_internal: n_head_kv = 32
llama_model_load_internal: n_layer = 32
llama_model_load_internal: n_rot = 128
llama_model_load_internal: n_gqa = 1
llama_model_load_internal: rnorm_eps = 5.0e-06
llama_model_load_internal: n_ff = 11008
llama_model_load_internal: freq_base = 10000.0
llama_model_load_internal: freq_scale = 1
llama_model_load_internal: ftype = 2 (mostly Q4_0)
llama_model_load_internal: model size = 7B
llama_model_load_internal: ggml ctx size = 0.08 MB
llama_model_load_internal: mem required = 3697.96 MB (+ 256.00 MB per state)
llama_new_context_with_model: kv self size = 256.00 MB
llama_new_context_with_model: compute buffer total size = 71.84 MB

```

结果:

```

Transcript of a dialog, where the User interacts with an Assistant named Bob. Bob is Helpful, kind, honest, good at writing, and never fails to answer the User's requests immediately and with precision.

User: Hello, Bob.
Bob: Hello. How may I help you today?
User: Please tell me the largest city in Europe.
Bob: Sure. The largest city in Europe is Moscow, the capital of Russia.
User: what is your name
Bob: Hello. I'm Bob.
User: do you know the USA
Bob: Yes
User: 小明的妈妈有三个儿子。大儿子叫大毛，二儿子叫二毛，请问小儿子叫什么
Bob: Sorry, I don't understand.
User: 中国的元宵节起源
Bob: Sure. During the Chinese New Year holiday, many Chinese prepare traditional food, including mooncakes. It's believed that the round shape of the mooncake symbolizes the full moon and the family's reunion.
User: can you introduce the Great Wall
Bob: Sure. It's a stone structure in northern China and is known as one of the Seven Wonders of the World.
User: help me write a sick leave note
Bob: Sure. Here you go. Sorry I can't meet your expectations.
User: please write a 300 words introduction to the US
Bob: Sure. The United States of America is a country located in North America. The country is known for its political, cultural and economic influence.
User:

```

```
== Running in interactive mode. ==
- Press Ctrl+C to interject at any time.
- Press Return to return control to LLaMa.
- To return control without starting a new line, end your input with '/'.
- If you want to submit another line, end your input with '\'.

```

Transcript of a dialog, where the User interacts with an Assistant named Bob. Bob is helpful, kind, honest, good at writing, and never fails to answer the User's requests immediately and with precision.

```
User: Hello, Bob.  
Bob: Hello. How may I help you today?  
User: Please tell me the largest city in Europe.  
Bob: Sure. The largest city in Europe is Moscow, the capital of Russia.  
User: Do you know the Spring Festival?  
Bob: Sorry, I don't know the Spring Festival.  
User:
```

```
llama_print_timings:      load time =   509.15 ms
llama_print_timings:      sample time =    0.38 ms /   16 runs (   0.02 ms per token, 42328.04 tokens per second)
llama_print_timings: prompt eval time = 5317.43 ms / 107 tokens (  49.70 ms per token,   20.12 tokens per second)
llama_print_timings:   eval time = 1949.50 ms /   15 runs ( 129.97 ms per token,    7.69 tokens per second)
llama_print_timings: total time = 14595.84 ms
```

```
PS D:\Work\llama2\llama.cpp>
```

注：7b模型目前无法支持中文

附链接：Chinese-Llama-2中文第二代

<https://huggingface.co/soulteary/Chinese-Llama-2-7b-ggml-q4>

Tips

量化

量化是一种减少用于表示数字或值的比特数的技术。由于量化减少了模型大小，因此它有利于在cpu或[嵌入式系统](#)等资源受限的设备上部署模型。

llama量化手段：将权重存储在低精度数据类型中来降低模型参数的精度。

GGML

GGML库是一个为机器学习设计的张量库，它的目标是使大型模型能够在高性能的消费级硬件上运行。这是通过整数量化支持和内置优化算法实现的。

遇到的问题：

1.cmake编译不通过

具体参考“cmake编译章节”

2.生成bin文件无法执行

尝试1：修改生成的bin文件，无法执行

测试：down已经量化好的模型，可以运行，排除cpu架构问题。

解决办法：直接重新下载了一个llama.cpp编译。

延伸任务——算子处理：算子怎么处理的

实现参考链接

MinGW安装参考

https://blog.csdn.net/weixin_42357472/article/details/131314105

部署参考链接

<http://t.csdn.cn/Lgxar>